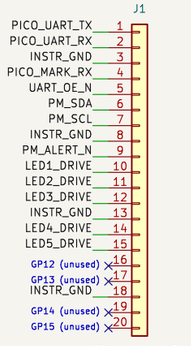
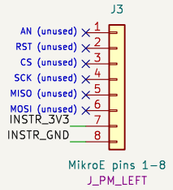
CARRIER — instrument side (generated by gen_carrier_sch.py; do not edit by hand). Everything here is powered by the Pico's own USB; nothing is fed from the DUT.

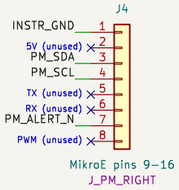
INSTR_3V3 comes only from Pico pin 36 (3V3_OUT). Pico VBUS (40) and VSYS (39) are left unconnected. Pico, MikroE and microSD plug into Harwin sockets J1–J5.

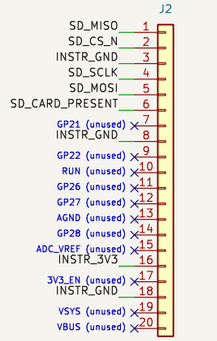
J2 pin k = Pico pin 20+k; J4 pin k = MikroE pin 8+k. Physical socket order is fixed at footprint stage (Pico pin 21 is opposite pin 20).

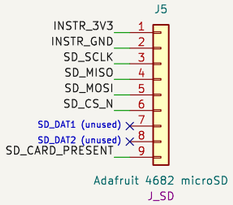
Sense: J7 -> C1 (100n across the Kelvin pair, close to the module) -> J6 wire pads -> MikroE IN1+ / IN1− / IN2 VBUS screw terminals. J7 pin 4 is the shared ground reference.

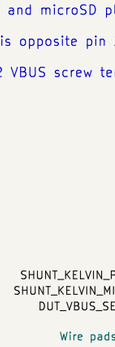


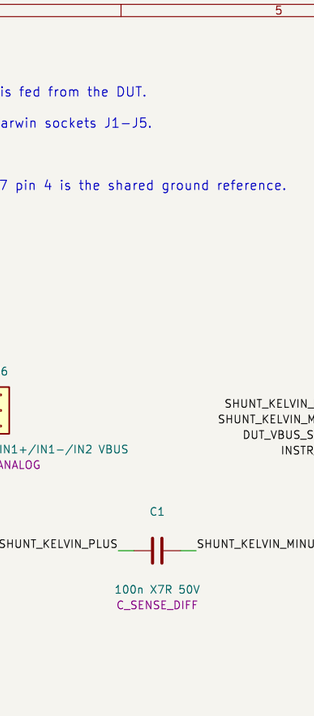


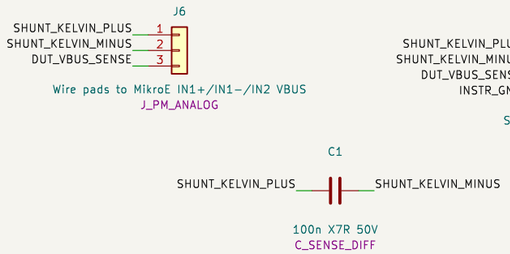


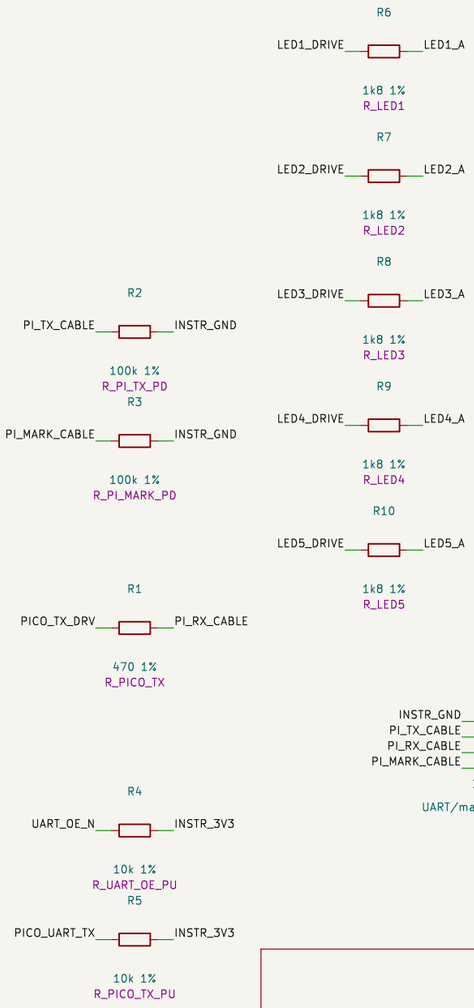




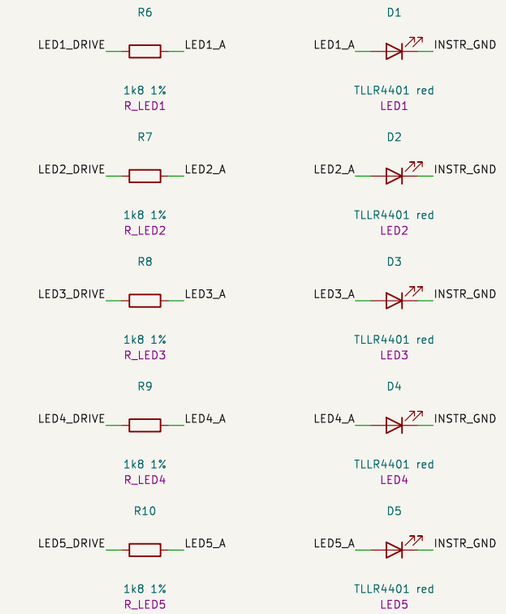


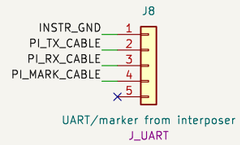






Indicators: Pico GP7–GP11 -> 1.8k -> red LED -> ground.





UART (docs/uart-interface-qualification.md): U1 is USB-powered. Keep UART_OE_N high (R4) until the Pico has set up its pins. Pico TX is normal polarity (R5 idle-high).

R2/R3 load the Pi outputs (approved UART-bias exception). R1 is the 470R source resistor for Pico->Pi. U1 channel 4 unused: input low, enable high.

Prototype — not released

picopower-proto

Sheet: /

File: carrier.kicad_sch

Title: Carrier

Size: A3	Date:	**Rev:**
KiCad E.D.A. 9.0.9		Id: 1/1